Chan Sherwin Stephen

Singapore | sherwin0402@gmail.com | sherwin0402.github.io | sherwin-chan

About Me

I am a final-year Ph.D. student at NTU, Singapore, advised by Prof. Ang Wei Tech, specializing in physics-based simulation of physical human-robot interaction for assistive and rehabilitative devices. I use MuJoCo as my primary platform to build realistic digital twins with skeletal, musculoskeletal, and soft-body models and design Real2Sim2Real frameworks for personalised robotic controller. My work bridges simulation-driven machine learning afor safe and intuitive HRI strategies to develop intelligent, physically interactive systems. I am currently exploring the use of foundation models and large language models to improve the fidelity of human-robot interaction in simulation.

Education

Nanyang Technological University, Singapore — Sept 2021 – Aug 2025 (Expected)

Ph.D. in Mechanical Engineering

- **Supervisor:** Prof Ang Wei Tech
- **Proposed Dissertation:** Accelerating the Development of Assistive Robotics through Accurate Human-In-The-Loop Robotic Simulation of Physical Human-Robot Interaction
- **Research Interests:** Physics-Based Simulation, Human-Robot Interaction, Machine Learning, Foundation Models, Large Language Models

Nanyang Technological University, Singapore — Aug 2017 – May 2021

BE in Mechanical Engineering with a Specialization in Robotics and Mechatronics

- **GPA:** 4.84/5.00, Dean's List for 2017/2018
- **Awards:** Nanyang Scholarship

Research Projects

Human-In-The-Loop Robotic Simulator — Sept 2021 – present

A simulation framework for physical human-robot interaction (pHRI) investigation to personalise robotic systems.

- Developed a human-in-the-loop simulation framework for assistive robotics, featuring personalized digital twins with varying abilities and disabilities. The pipeline incorporates skeletal, musculoskeletal, and soft body models, using reinforcement learning to simulate diverse human capabilities and enable realistic, adaptive testing of robotic systems.
- Designed and validated a Real2Sim2Real framework to personalize robotic controllers based on individual user characteristics, enabling improved pHRI through simulation-informed adaptation and real-world testing.
- Investigating various human-robot interaction modalities in simulation, including soft body dynamics, mass-spring-damper models, and other methods to accurately represent physical interactions.
- Simulated a range of assistive robotic systems - including robot-assisted feeding arms, lower limb and upper limb exoskeletons, gait assistive robots, using MuJoCo.

Mobile Third Arm Robot — 2020 – 2022

Vision-guided mobile third arm co-bot for patient transfers between bed and wheelchair.

- Developed a mobile third arm robot that assists caregivers for moderate assisted pivot transfer between the bed and wheelchair.
- Implemented a human-tracking computer vision algorithm to perform user following
- Developed human-robot interaction strategies for safe and intuitive operation of the mobile third arm robot between caregiver, patient and mobile third arm robot.

Design and Analysis of Underwater Robotic Systems — 2018 – 2019

3D printed underwater robotic manipulator for object grasping.

- Designed and 3D printed a robotic manipulator for an autonomous underwater vehicle to grasp objects from the pool floor.
- Created a bio-inspired flexible hull structure that deforms with depth to enhance rigidity while reducing weight.
- Conducted finite element analysis to optimize both manipulator and hull designs for strength, hydrodynamic performance, and operational depth.

Work Experience

Project Officer, Rehabilitation Research Institute of Singapore – Singapore — Aug 2021 – present

- Spearheaded the Human-In-The-Loop Robotic Simulator project, focusing on developing a platform that combines realistic human digital twins with assistive robots for human-in-the-loop simulations.
- Mentored seven undergraduate students on final year projects, guiding them through the research process and providing technical support.
- Assisted in preparing funding proposals to support current and future projects of the institute.

Engineering Intern, Oishii – New Jersey, USA — June 2019 – June 2020

- Designed an indoor vertical farm structure accommodating 250,000 plants with a novel rack system that increases efficiency of construction and assembly process by 90%.
- Pioneered development of a USD 400,000 R&D facility consisting of ten independent growing environments tracking more than 80 plant parameters to determine optimal plant growth and yield.
- Led an engineering team of 12 to implement testing and troubleshooting schedule to all farm sub-systems as lead test engineer to bring the world's largest indoor strawberry vertical farm facility operational.
- Evaluated 32 engineering candidates for engineering roles, mentored 19 new engineering technicians to improve mechanical and electrical skillsets and align with company working norms.

Publications

ORBiT: Optimizing Robot-Assisted Bite Transfer Leveraging a Real2Sim2Real Framework — Oct 2025
Sherwin Stephen Chan, J Anne Yow, Yi Heng San, Vasanthamaran Ravichandram, Yifan Wang, Lek Syn Lim, Wei Tech Ang
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

A Human-In-The-Loop Simulation Framework for Evaluating Control Strategies in Gait Assistive Robots — May 2025
Yifan Wang†, ***Sherwin Stephen Chan†***, Mingyuan Lei, Lek Syn Lim, Henry Johan, Bingran Zuo, Wei Tech Ang
IEEE International Conference on Robotics and Automation (ICRA), † Denotes equal contribution

Simulating Safe Bite Transfer in Robot-Assisted Feeding with a Soft Head and Articulated Jaw — May 2025
Yi Heng San, Vasanthamaran Ravichandram, J Anne Yow, ***Sherwin Stephen Chan***, Yifan Wang, Wei Tech Ang
IEEE International Conference on Rehabilitation Robotics (ICORR)

Personalised 3D Human Digital Twin with Soft-Body Feet for Walking Simulation — Dec 2024
Kum Yew Loke, ***Sherwin Stephen Chan***, Mingyuan Lei, Henry Johan, Bingran Zuo, Wei Tech Ang
International Conference on Social Robotics

Creation and evaluation of human models with varied walking ability from motion capture for assistive device development — Sept 2023
Sherwin Stephen Chan, Mingyuan Lei, Henry Johan, Wei Tech Ang
IEEE International Conference on Rehabilitation Robotics (ICORR)

Investigation of Modeling Differences between OpenSim and Visual3D for Gait Analysis of Healthy Gait — Aug 2023
Beth Eng Wan Xuan, ***Sherwin Stephen Chan***, Henry Johan, Lek Syn Lim, Bingran Zuo, Wei Tech Ang
International Convention on Rehabilitation Engineering and Assistive Technology

Skills

Programming Languages: Python, C++, C

Robotics: MuJoCo, OpenSim, Machine Learning, Reinforcement Learning, IsaacLabs, ROS, SolidWorks, SolidEdge, STM32, ESP32, Linux

Languages: English, Chinese, Hokkien (Dialect), Tagalog